

Daily Work Report

Employee Name: Atharva Ravikiran Avhad

Date: 27/07/2026

Reporting Manager: Aditya Chauha

Work Summary

Focused on Days 3 and 4 of the Week 3 objective. Developed and executed an automated benchmarking suite (benchmark.py) to process multiple PDF categories and classify extraction failures. Additionally, implemented a performance profiling script (profile_app.py) using cProfile and memory_profiler to identify CPU and RAM bottlenecks in the PDF parsing layer. Based on the metrics, verified that our asynchronous threading optimizations (from Day 2) correctly handle heavy multi-page financial documents without blocking.

Completed Tasks

- **Benchmarking Suite (Day 3):** Created benchmark.py which feeds a directory of categorized PDFs (Financial, Legal, Scanned, etc.) into the API. Automatically records extraction time, success rate, and HTTP status codes into a benchmark_report.json file.
- **Failure Analysis:** Extracted logs from the benchmark script to categorize failure root causes (e.g., classifying structural failures vs. corrupted document errors, validating our custom PDFCorruptedException handlers).
- **Performance Profiling (Day 4):** Developed profile_app.py to trace function-level execution time and memory consumption during the extraction of heavy PDFs (like q3_report.pdf).
- **Optimization Validation:** Confirmed that the hybrid engine (which runs Camelot and PyMuPDF in a ThreadPoolExecutor) successfully reduces overall latency compared to sequential processing.

Pending Tasks

- Prepare the final, comprehensive Engineering Report summarizing the benchmarking and profiling results.
- Complete the Production Readiness Checklist.
- Finalize Docker containerization (if required for deployment).

Plan for Tomorrow

- Compile the data from benchmark_report.json and the cProfile outputs into a formal Engineering Report.
- Cross-check the Production Readiness Checklist to ensure all environment variables, health checks, and CI/CD pipelines are documented.
- Wrap up Week 3 objectives and present the completed microservice.

Employee Declaration

I hereby confirm that the above information accurately represents the work completed by me on the mentioned date.

Employee Signature:

Atharva Avhad